

SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX lines 3522–3530; original printed pp. 119–120, physical source-PDF pp. 103–104, and recomposed running pp. 95–96. The `\pageoriginale` marker occurs inside line 3530, so none of these locator systems is collapsed into another. Blank line 3531 and Remark 3.7 at line 3532 are excluded. The raw continuation cursor is line 3531 and the next substantive cursor is line 3532. The Latin-1 French authority remains byte-exact. Same-edition PDFs are manifestation/layout evidence only; the external English candidate is comparison-only.

Proof of Lemma 3.6. Observe that the hypothesis $\mathcal{O}_X \simeq i_*\mathcal{O}_U$ is preserved under flat base extension, since i is quasi-compact and, in this case, formation of the direct image commutes with inverse image. This hypothesis implies that the functor

$$\mathbf{Et}(V) \longrightarrow \mathbf{Et}(U \cap V)$$

defined by

$$V' \longmapsto V' \times_V (V \cap U)$$

is fully faithful, as follows from the interpretation of an étale covering in terms of a coherent locally free algebra. It remains to prove effectivity. One may, for example, form the fiber square X_2 and the fiber cube X_3 of X_1 over X , and observe that a faithfully flat and quasi-compact morphism is a morphism of universal effective descent for the fibered category of étale coverings over the category of preschemes. The conclusion then follows formally^(*).

^(*)Cf. J. Giraud, *Méthode de la descente*, Mémoire no. 2 du *Bulletin de la Société Mathématique de France* (1964).